Individual assignment 5

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2026-05-10

Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(janitor)
library(readxl)
library(snakecase)
library(scales)

# file organization
library(here)

# plots
library(patchwork)
library(viridis)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                    sheet = "Aquatic Insects")
```

```r
# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))
```

Cleaning data

```r
# creating a new object from sites
ncos_sites <- sites |>
  # filtering to only include NCOS quarterly sampling
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

```r
# creating a new object
water_quality_clean <- water_quality |>
  # cleaning column names
  clean_names() |>
  # use a filtering join, only include sites in ncos_sites
  semi_join(ncos_sites,
            by = "site") |>
  # creating new column to join with later data frames
  unite("date_site", date, site,
        remove = TRUE) |>
  # grouping by date/site
  group_by(date_site) |>
  # caluclating median, pH, salinity, and DO
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungrouping data frame
  ungroup() |>
  # filtering out outliers
  filter(date_site != "2022-08-12_NVBR")
```

```r
# create a new object from taxon_list
taxon_list_clean <- taxon_list |>
  # change names in verbatim_name to snake case
```

```r
  mutate(verbatim_name = to_any_case(verbatim_name,
                                     case = "snake"))
```

```r
# create a new object from aq_ins
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filter join to only include sites matching with ncos_sites
  semi_join(ncos_sites,
            by = "site") |>
  # rename column date_on_vial to date
  rename(date = date_on_vial) |>
  # select columns of interest
  select(date, sample_type, site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
         diptera_chironomid,
         annelida_polychaete) |>
  # filter to only include FB250/FB 250 samples
  filter(sample_type %in% c("FB250", "FB 250")) |>
  # pivot data frame
  pivot_longer(
    # columns to make longer
    cols = ostracod:annelida_polychaete,
    # new column for the old column names
    names_to = "taxon",
    # new column for the values in each of the old columns
    values_to = "abundance"
  ) |>
  # group by date, site, and taxon
  group_by(date, site, taxon) |>
  # summarize to find average abundance per liter
  summarize(ave_lit = sum(abundance, na.rm = TRUE) / 7.5) |>
  # ungroup
  ungroup() |>
  # combine date and site into date_site, don't remove date and site columns
  unite("date_site", date, site, remove = FALSE) |>
```

```
  # left join water_quality_clean by date_site column
  left_join(water_quality_clean, by = "date_site") |>
  # left join taxon_list_clean by taxon(aq_ins)/verbatim_name(taxon_list) column
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
  # mutate to create a column for full names of site codes
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "South of Venoco Bridge",
    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NWP" ~ "West Pond",
    site == "NDC" ~ "Devereux Creek"
  ))
```

```
# creating new object from clean aquatic inverts
copepods <- aq_ins_clean |>
  # filter to only include copepods
  filter(taxon == "copepod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))
```

Visualizations from class

```
# base layer
salinity <- ggplot(data = aq_ins_clean,
       # x-axis: site
       # y-axis: mean salinity
       mapping = aes(x = site_full,
                     y = med_sal)) +
  # first layer: salinity measurements at each survey
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(width = 15)) +
  # second layer: points to represent means
  stat_summary(fun = median,
```

```
               color = "red",
               size = 1) +
  # relabelling x- and y-axes
  labs(x = "Site",
       y = "Median salinity (ppt)",
       title = "Site salinity") +
  # different theme
  theme_bw()
```

```
# base layer
copepods <- ggplot(data = copepods,
                   # x-axis: site
                   # y-axis: mean abundance/L
                   mapping = aes(x = site_full,
                                 y = log_ave_lit)) +
  # first layer: points to represent observations
  geom_jitter(height = 0,
              shape = 21,
              width = 0.1) +
  # second layer: points to represent medians
  stat_summary(geom = "point",
               fun = "median",
               size = 4,
               color = "blue",
               shape = 17) +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(10)) +
  # relabelling x- and y-axis, adding title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Copepods") +
  # different theme
  theme_bw()
```

1. Visualizing differences across sites in major taxon abundance

a. Plan your figure

In part 1d, you should end up with a three panel figure. All three panels should have site on the x-axis, but each panel will have a different y-axis.

In part 1c, you will create one component of the three panel figure. Provide a list of the axes and geometries you will need to create a figure in 1c showing differences between sites in ostracod abundance.

- x-axis: `site_full`
- y-axis: `log_ave_lit`
- geometry to represent average abundance/L: `geom_jitter`
- geometry to represent medians: `stat_summary`

b. Wrangle your data

The `aq_ins_clean` data frame will need some further wrangling before you create the figure you described in part 1a.

In this section, create the data frame you will need to make your figure.

At the end of your code chunk, display 10 random rows from your data frame using `slice_sample()`.

```
# create a new object from aq_ins_clean
ostracods <- aq_ins_clean |>
  # filter to only include ostracod taxon
  filter(taxon == "ostracod") |>
  # log transform average abundance/L as a new column
  mutate(log_ave_lit = log(ave_lit + 1))

# display 10 rows from ostracods
slice_sample(ostracods, n = 10)
```

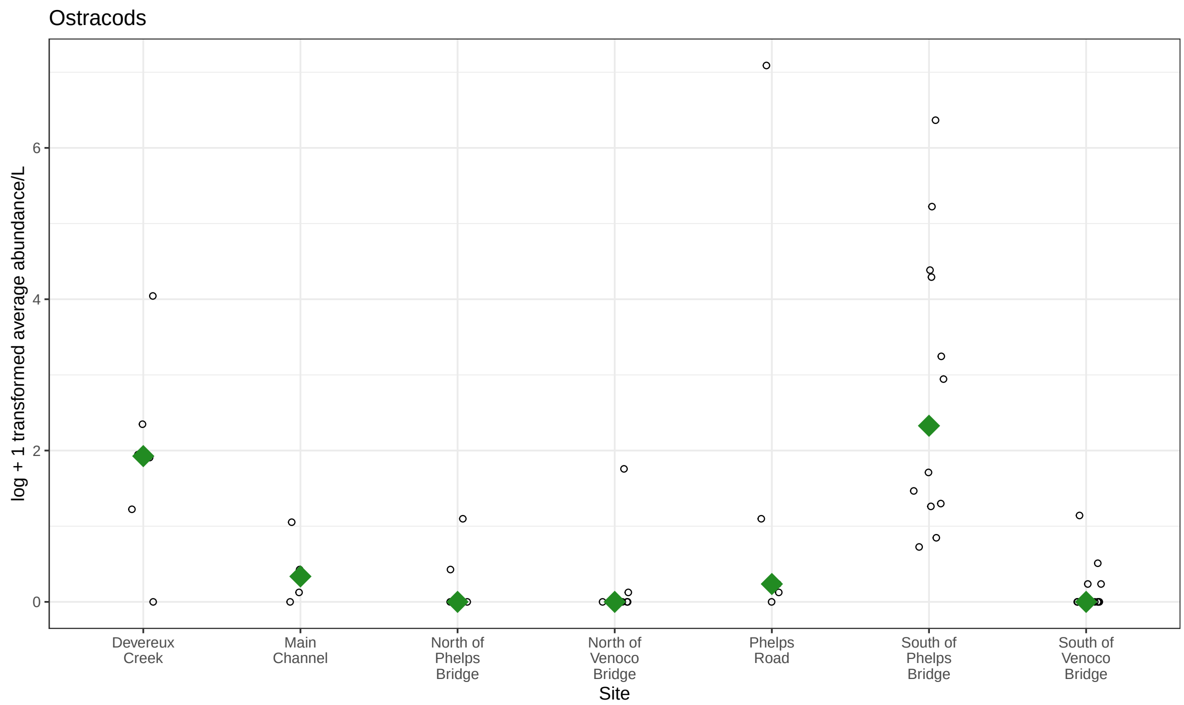
```
# A tibble: 10 x 24
   date_site       date                site  taxon ave_lit med_ph med_sal med_do
   <chr>           <dttm>              <chr> <chr>   <dbl>  <dbl>   <dbl>  <dbl>
 1 2023-11-11_NEC  2023-11-11 00:00:00 NEC   ostr~   0      NA      38.6    0.25
 2 2021-04-15_NPB  2021-04-15 00:00:00 NPB   ostr~ 185.      8.09    3.57   8.24
 3 2023-08-12_NEC  2023-08-12 00:00:00 NEC   ostr~   0      NA      NA     NA
 4 2022-08-15_NMC  2022-08-15 00:00:00 NMC   ostr~   0.133   7.61   NA     NA
 5 2023-11-12_NMC  2023-11-12 00:00:00 NMC   ostr~   0.533  NA      47     14.0
 6 2022-01-28_NDC  2022-01-28 00:00:00 NDC   ostr~   9.47    7.65    2.00   9.55
 7 2023-05-05_NPB  2023-05-05 00:00:00 NPB   ostr~  24.7    NA      NA     12.1
 8 2021-09-01_NPB2 2021-09-01 00:00:00 NPB2  ostr~   0      NA      NA     NA
 9 2022-11-12_NEC  2022-11-12 00:00:00 NEC   ostr~   0       7.33    7.01   2.4
10 2022-05-12_NPB  2022-05-12 00:00:00 NPB   ostr~   1.07    7.97   11.4    8.17
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
```

```
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <chr>, log_ave_lit <dbl>
```

c. Code your figure

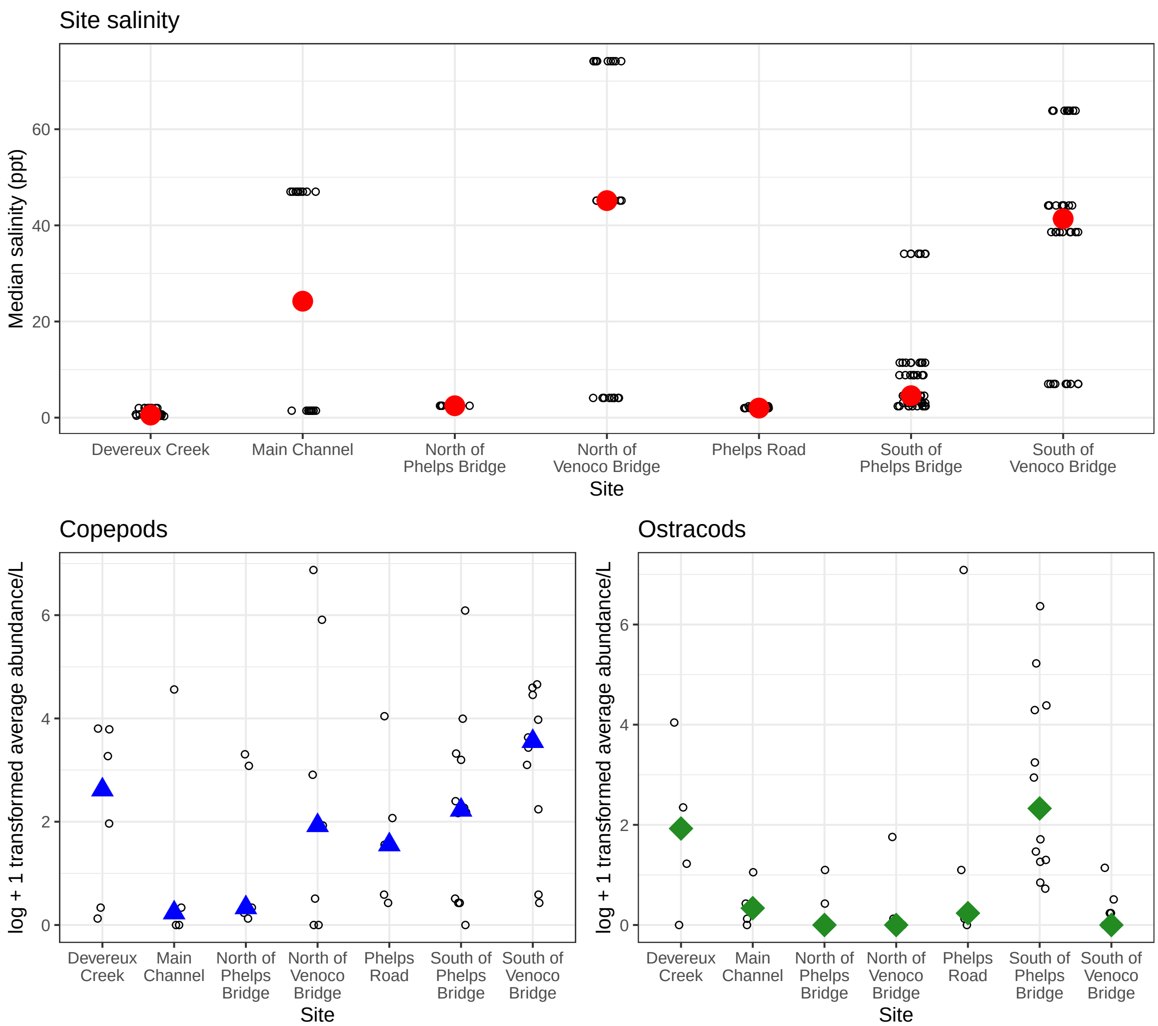
```
# base layer
ostracods <- ggplot(data = ostracods,
                   # x-axis: site
                   # y-axis: mean abundance/L
                   mapping = aes(x = site_full,
                                 y = log_ave_lit)) +
  # first layer: points to represent observations
  geom_jitter(height = 0,
              shape = 21,
              width = 0.1) +
  # second layer: points to represent medians
  stat_summary(geom = "point",
               fun = "median",
               size = 6,
               color = "forestgreen",
               shape = 18) +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(10)) +
  # relabelling x- and y-axis, adding title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Ostracods") +
  # different theme
  theme_bw()

# displaying object
ostracods
```



d. Create a multipanel figure

```
# stack salinity on top and copepods and ostracods side by side
salinity / (copepods + ostracods)
```



e. Interpret your figure

In 1-2 sentences each, describe:

- differences between sites in copepod and ostracod abundance
  - On average, there appears to be a higher abundance of copepods across sites compared to ostracods. We can visually tell because of how the blue points representing the central tendency of abundance are generally higher in the y-axis direction in the copepod plot than in the ostracod plot.
- how ostracod abundance may relate (or not) to site-level differences in salinity

- There doesn't appear to be a clear relationship between ostracod abundance and salinity, which we can visually tell because the positions of the median points for ostracod abundance don't appear any higher or lower in response to site salinity (they don't consistently "move" with each other). We might otherwise expect an inverse correlation, but sites like North of Phelps and Phelps Road disrupt this idea.

2. Visualizing total abundance as a function of salinity

a. Plan your figure

At the end of part 2c, you should have a figure:

- depicting log-transformed average abundance/L on the y-axis and salinity on the x-axis
- with different panels for each taxon in aq_ins_clean with free scales for both axes
- with each taxon in its own color

Provide a list of values, geometries, and/or arguments that would be appropriate to make a figure matching this description.

- x-axis: `med_sal`
- y-axis: `log_ave_lit`
- color: `taxon`
- geometry: `geom_point()`
- facets: `taxon`

b. Wrangle your data

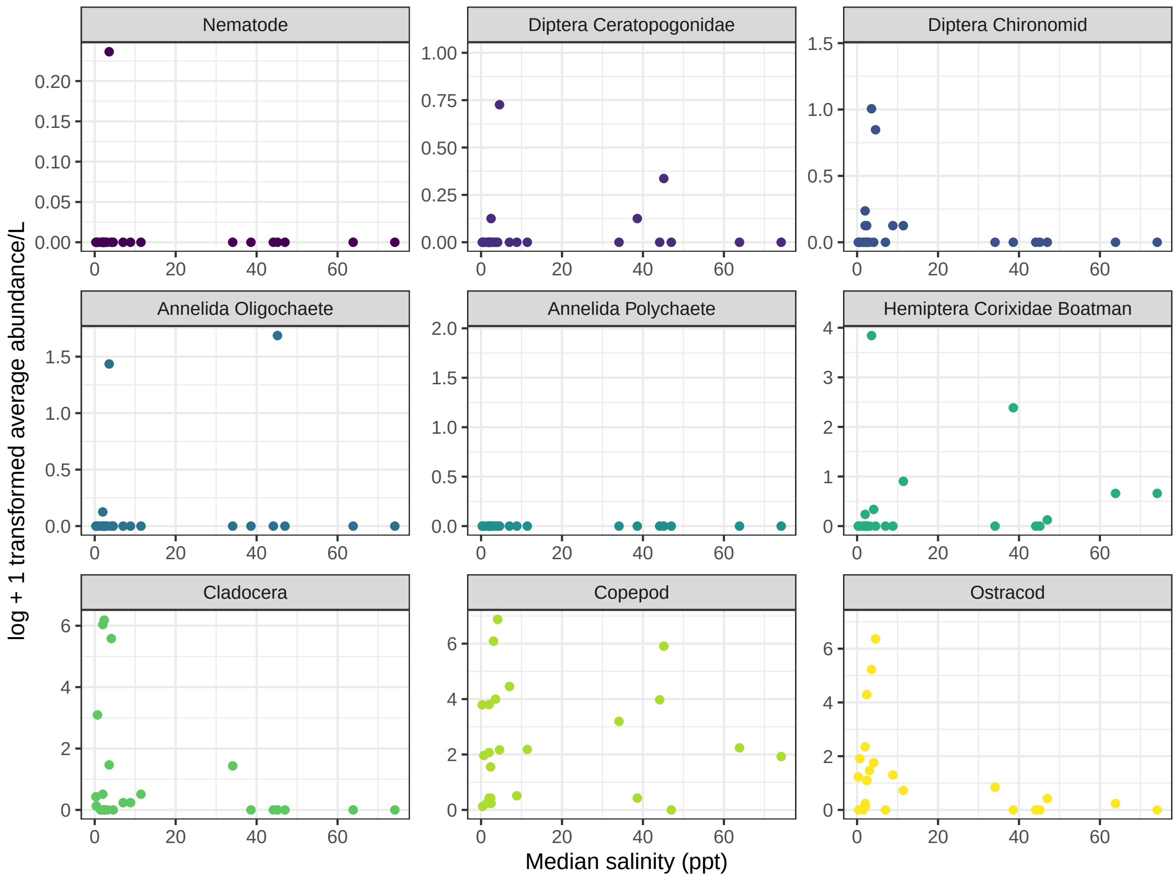
```r
# create a new object from aq_ins_clean
aq_ins_salinity <- aq_ins_clean |>
  # log transform average abundance/L as a new column
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  # set taxon to title case and reorder by max abundance
  mutate(taxon_full = to_any_case(taxon, case = "title"),
         taxon_full = as_vector(taxon_full),
         taxon_full = fct_reorder(taxon_full, log_ave_lit, max)) |>
  # arrange data frame by taxon
  arrange(desc(taxon_full)) |>
  # select columns of interest
  select(med_sal, log_ave_lit, taxon_full)
```

```r
# display 10 rows from aq_ins_salinity
slice_sample(aq_ins_salinity, n = 10)
```

```
# A tibble: 10 x 3
   med_sal log_ave_lit taxon_full
     <dbl>       <dbl> <fct>
 1  NA           0     Nematode
 2   2.47        0.125 Diptera Ceratopogonidae
 3   0.688       1.96  Copepod
 4  NA           0     Diptera Ceratopogonidae
 5  NA           0     Diptera Ceratopogonidae
 6   3.08        1.47  Ostracod
 7  NA           0     Hemiptera Corixidae Boatman
 8  NA           0     Hemiptera Corixidae Boatman
 9  NA           0     Annelida Oligochaete
10  NA           3.25  Ostracod
```

c. Code your figure

```r
# base layer: ggplot
ggplot(data = aq_ins_salinity,
       # set x- and y-axes, differentiate color by taxon
       aes(x = med_sal,
           y = log_ave_lit,
           color = taxon_full)) +
  # first layer: point geometry
  geom_point() +
  # create panels by taxon, set free scale y-axis
  facet_wrap(~ taxon_full,
             scales = "free") +
  # add labels to axes
  labs(x = "Median salinity (ppt)",
       y = "log + 1 transformed average abundance/L") +
  # change colors
  scale_color_manual(values = viridis(9)) +
  # change theme
  theme_bw() +
  # remove legend
  theme(legend.position = "none")
```



d. Interpret your figure

Across all taxa, there appears to be a general negative relationship between average abundance and salinity. On average, as we move along the x-axis, the points tend to be positioned lower along the y-axis. This relationship is weaker with taxa like the Copepod and Annelida Polychaete, and stronger with taxa like the Ostracod and Cladocera.